

Secure SDN Traffic Management using Post-Quantum Cryptography

Mentor Name: Dr. Abhishek Narwaria

Overview

The rapid growth of enterprise, campus, and mission-critical communication networks demands intelligent, scalable, and secure network management solutions. Traditional networking architectures suffer from limited flexibility and are increasingly vulnerable to cyber threats, especially with the emergence of quantum computing that threatens classical cryptographic systems. Software Defined Networking (SDN) provides centralized and programmable network management, while Post-Quantum Cryptography (PQC) offers quantum-resistant security mechanisms for future communication systems. This internship focuses on developing a lightweight SDN-based secure traffic management framework integrated with basic PQC concepts. The aim is to provide hands-on exposure to SDN controllers, OpenFlow communication, network traffic monitoring, and secure controller-switch communication using post-quantum techniques.

The internship will focus on the following objectives:

(a) To study SDN architecture, OpenFlow communication, and programmable networking concepts. (b) To set up and configure SDN simulation environments using Mininet and SDN controllers such as Ryu or Floodlight. (c) To implement traffic monitoring and QoS-based routing mechanisms for efficient traffic management. (d) To explore lightweight PQC algorithms and analyze their suitability for SDN environments. (e) To perform comparative performance analysis of secure and non-secure SDN communication using metrics such as latency, throughput, and packet delivery ratio.

The project scope includes:

(a) Development of SDN-based traffic visualization and monitoring modules. (b) Simulation of controller-switch communication under different traffic conditions. (c) Experimental evaluation using Mininet, Wireshark, and network analysis tools. (d) Comparative analysis of traditional and PQC-enabled secure communication approaches.

We expect interns to dedicate 2–3 days per week (8-hour workdays) as follows:

1. **Weeks 1 and 2:** Literature review on SDN, OpenFlow, and Post-Quantum Cryptography along with installation and configuration of SDN tools.
2. **Weeks 3 and 4:** Development of SDN traffic monitoring and routing modules with basic secure communication implementation.
3. **Weeks 5 and 6:** Performance evaluation, benchmarking, report preparation, and presentation of results.

Guidance Approach

As the research mentor, the following activities will be conducted:

- (a) Providing curated reading materials related to SDN, OpenFlow, and PQC.
- (b) Conducting weekly mentoring sessions on SDN implementation, traffic analysis, and technical documentation.
- (c) Holding weekly progress meetings to resolve technical challenges and monitor implementation progress.
- (d) Providing structured feedback on coding, documentation, and experimental results.